ROUTING AND ROUTING PROTOCOLS

1.Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.

ANS:

Network Diagram: Three Devices Connected Through a Router

Wi-Fi Router

LAN IP: 192.168.1.1

WAN IP: 203.0.113.2

Wi-Fi / Ethernet connections

My Phone

Wi-Fi

IP: 192.168.1.10

Friend's Laptop

Wi-Fi

IP: 192.168.1.20

Smart TV

Ethernet

IP: 192.168.1.30

IP Address Table

Device / Interface	**IP Address**	**Connection**
Router LAN interface	192.168.1.1	Local network gateway
Router WAN interface	203.0.113.2	Internet-facing interface (example)
My Phone	192.168.1.10	Wi-Fi
Friend's Laptop	192.168.1.20	Wi-Fi
Smart TV	192.168.1.30	Ethernet cable

**2.Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made.

Hint: Use made-up IP addresses and show the routing table entries that would be checked at each hop.**

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Manual IP Routing Simulation

1. Network Scenario

A laptop sends a packet to a friend's computer located in a different subnet. Two routers are used to forward the packet.

My Laptop

IP: 192.168.1.10/24

Router 1

192.168.1.1 → 10.0.0.1

Router 2

10.0.0.2 → 192.168.2.1

Friend's Computer

IP: 192.168.2.20/24

2. Step-by-Step Packet Travel

1. My laptop checks the destination. It compares 192.168.2.20 with its own subnet, 192.168.1.0/24. Since the destination is in a different subnet, it sends the packet to its default gateway, 192.168.1.1.

2. Router 1 checks its routing table. It finds a route to 192.168.2.0/24 through Router 2, using next-hop address 10.0.0.2.

3. Router 2 receives the packet. It checks its routing table and finds that 192.168.2.0/24 is directly connected to its local interface.

4. Router 2 forwards the packet. It sends the packet to the friend's computer at 192.168.2.20.

5. The friend's computer receives the packet. The packet has reached its destination. Reply traffic follows the appropriate routes back to the laptop.

3. Routing Table Entries

Router 1 routing table

Destination Network	Next Hop	Interface
192.168.1.0/24	Directly connected	LAN
10.0.0.0/30	Directly connected	Link to Router 2

192.168.2.0/24	10.0.0.2	Link to Router 2

Router 2 routing table

Destination Network	Next Hop	Interface
10.0.0.0/30	Directly connected	Link to Router 1
192.168.2.0/24	Directly connected	LAN
192.168.1.0/24	10.0.0.1	Link to Router 1

4. How the Routing Decision Is Made

Routers examine the destination IP address and choose the most specific matching route in their routing tables. This is called longest-prefix matching. If the destination network is directly connected, the router forwards the packet on that interface; otherwise, it uses the selected next hop.

3.Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

ANS:

Routing Table for a Zomato-Style Delivery Network

A fictional food delivery company has three branches in different cities: Ahmedabad, Mumbai, and Delhi. Each branch uses a different IP subnet, and routers forward packets between the branches.

1. Network Details

Branch	City	Network Address	Subnet Mask
Branch 1	Ahmedabad	192.168.10.0	255.255.255.0
Branch 2	Mumbai	192.168.20.0	255.255.255.0
Branch 3	Delhi	192.168.30.0	255.255.255.0

2. Routing Table

Assume the Ahmedabad router is the main router, connected to the other two branches through separate router links.

Destination Network	Subnet Mask	Next Hop	Interface
192.168.10.0	255.255.255.0	Directly connected	G0/0
192.168.20.0	255.255.255.0	10.0.0.2	G0/1

192.168.30.0 255.255.255.0 10.0.0.6 G0/2

3. Explanation

- Network Address: Identifies the destination branch network.
- Subnet Mask: Defines the network portion of the IP address.
- Next Hop: Specifies the IP address of the next router to receive the packet.
- Interface: Specifies the outgoing router interface used to forward the packet.

Example: When a delivery employee in Ahmedabad sends data to a computer in Mumbai, the Ahmedabad router checks its routing table and forwards the packet to next-hop address 10.0.0.2 through interface G0/1.

4.Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

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Comparison of Static Routing and Dynamic Routing

Static routing uses routes configured manually by a network administrator, while dynamic routing uses protocols such as RIP and OSPF to learn and update routes automatically.

Type	**Pros (Advantages)**	**Cons (Disadvantages)**
Static Routing	1. Simple for small networks, such as a small restaurant's office network.	1. Routes must be updated manually when the network changes.
	2. Uses fewer router resources because no routing protocol updates are needed.	2. If a link fails, traffic will not automatically switch to another route unless a backup is configured.
	3. Provides predictable paths and greater administrative control.	3. Difficult to manage across many branches of a company like Zomato.
Dynamic Routing (RIP/OSPF)	1. Automatically learns routes between different branch networks.	1. Uses additional CPU, memory, and bandwidth for routing updates.
	2. Can find alternative routes when a network link fails.	2. More complex to configure and troubleshoot.
	3. Suitable for large networks connecting multiple cities and offices.	3. Incorrect configurations or routing updates can cause network problems.

Real-World Examples

- Static routing: A small restaurant uses a fixed route to connect its billing computer to a local server.
- Dynamic routing: A food delivery company like Zomato connects offices across Ahmedabad, Mumbai, and Delhi. OSPF can help routers learn routes and select alternatives if a network link fails.
- RIP: A small network can use RIP to exchange routes automatically, although its 15-hop limit makes it unsuitable for larger networks.

**5.Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does.

Hint: Paste the AI-generated config and annotate each part with a short description.**

ANS:

OSPF Routing Configuration Between Two Routers

Aim: To configure OSPF (Open Shortest Path First) routing between two routers so that they can exchange network information automatically.

1. Network Topology

Router 1

LAN: 192.168.1.0/24

Link IP: 10.0.0.1/30

OSPF Area 0

Router 2

Link IP: 10.0.0.2/30

LAN: 192.168.2.0/24

2. AI-Generated Configuration

The following example uses Cisco IOS commands.

Router 1 configuration

enable

configure terminal

router ospf 1

router-id 1.1.1.1

network 192.168.1.0 0.0.0.255 area 0

network 10.0.0.0 0.0.0.3 area 0

end

Explanation of each line

Command	**Purpose**
enable	Enters privileged EXEC mode.
configure terminal	Opens global configuration mode.
router ospf 1	Starts OSPF process 1.
router-id 1.1.1.1	Sets Router 1's unique OSPF identifier.
network 192.168.1.0 0.0.0.255 area 0	Enables OSPF on matching interfaces in the local LAN subnet.
network 10.0.0.0 0.0.0.3 area 0	Enables OSPF on the inter-router link subnet.
end	Returns to privileged EXEC mode.

Router 2 configuration

enable

configure terminal

router ospf 1

router-id 2.2.2.2

network 192.168.2.0 0.0.0.255 area 0

network 10.0.0.0 0.0.0.3 area 0

end

Explanation: Router 2 uses the same OSPF process and area, but has a different router ID. Its network commands enable OSPF for its own LAN and the shared inter-router link.

3. Verify OSPF

Run these commands on both routers:

show ip ospf neighbor

show ip route ospf

show ip protocols

- show ip ospf neighbor checks whether the routers have formed an OSPF neighbor relationship.
- show ip route ospf displays routes learned through OSPF.

- show ip protocols displays routing protocol settings.

4. Expected Result

If the interfaces are correctly configured, both routers can form an OSPF adjacency and learn routes to each other's LAN. For example, Router 1 should learn a route to 192.168.2.0/24 through Router 2.